

DD_FA-01 - Equipment Field Audit

Version: v1.0 **Bundle:** 10_other **Status:** Draft for review **Primary owner:** Field audit / operations quality team **Related DDs:** DD_WO-01, DD_OSR-01, DD_OSR-INSTANCE-01, DD_OSR-05, DD_OSR-RMA-01, DD_EM-02, DD_EM-CFG-03, DD_EM-CFG-04, DD_TCK-01, DD_REP-01, FOUNDATION_DROOLS

MVP simplification baseline: FA-01, FA-02, and FA-03 must be implemented as one field-audit capability in the first release, with `audit_type` driving type-specific rules, checklists, and routing. Do not create three services or three independent databases.

1. Purpose

FA-01 defines a controlled equipment field audit process for verifying that equipment recorded in SOPHIX exists in the expected field/customer location, has the expected serial identity, and is in an acceptable condition.

Use FA-01 when:

- operations wants to verify equipment installed at a customer premises;
- a contractor or field team must prove custody of assigned customer equipment;
- support suspects serial mismatch, damaged CPE, missing ONT/modem/STB, or wrong equipment location;
- internal audit needs evidence that equipment state in OSR-INSTANCE matches real-world field state;
- a discrepancy must be routed to support, recovery, RMA, write-off, or manual correction.

FA-01 is not the same as OSR-05 inventory audit. OSR-05 audits stock locations such as warehouse and contractor vans. FA-01 audits equipment in the field/customer context and uses technician/mobile evidence.

FA-01 is also not a replacement for normal WO completion evidence. Normal installation, shifting, support, and swap WOs already capture equipment bindings/photos in their own flows. FA-01 is used when the business needs a standalone audit campaign or an exception-driven verification after the normal job has already completed.

2. Scope And Boundaries

Capability	Owner
Audit campaign, audit task, expected vs observed comparison, discrepancy routing	FA-01
Field task assignment and technician execution when audit is WO-backed	WO-01
Current equipment instance state and customer/location binding	OSR-INSTANCE-01
Stock balances and location inventory	OSR-01
Warehouse/van inventory audit and stock adjustment/write-off governance	OSR-05
Equipment recovery, swap, RMA flow	OSR-RMA-01
Field user/contractor/team identity	EM-02
Permissions	EM-CFG-03
Approval for risky state correction/write-off	EM-CFG-04
Customer/service issue ticket	TCK-01
Audit reporting	REP-01
Configurable severity/routing rules	Drools

Boundary rules:

- FA-01 does not directly update OSR stock balances or equipment instance lifecycle.
- FA-01 routes approved corrections to OSR-INSTANCE, OSR-05, or OSR-RMA.
- FA-01 does not create purchase orders and does not call OSR-02 procurement.
- FA-01 can create a WO with `kind=FIELD_AUDIT` when technician scheduling/evidence is needed.

- If the audit is simple and internal, FA-01 may use lightweight audit tasks without WO, but mobile execution should still follow the same evidence standards.
- EM-CFG-04 is used for risky actions such as forced location correction, write-off, or marking unrecoverable equipment. Camunda is involved only when the approval policy mode is BPMN.

3. Process Overview

```

flowchart LR
    A["Backoffice creates audit campaign/task"] --> B["FA-01 loads expected equipment from OSR-INSTANCE"]
    B --> C["WO or audit task assigned to field user"]
    C --> D["Mobile app scans serial and captures evidence"]
    D --> E["FA-01 compares expected vs observed"]
    E --> F{"Discrepancy?"}
    F -- "No" --> G["Close audit task"]
    F -- "Yes" --> H["Open discrepancy"]
    H --> I["Drools selects severity and route"]
    I --> J["Create ticket, RMA/recovery, OSR correction, or approval"]

```

4. Status Models

4.1 Campaign Status

Status	Meaning
DRAFT	Campaign being prepared.
SCHEDULED	Campaign approved/scheduled.
IN_PROGRESS	Audit tasks are being executed.
RECONCILING	Submitted observations are being reviewed.
CLOSED	Campaign completed.
CANCELLED	Campaign cancelled.

4.2 Audit Task Status

Status	Meaning
CREATED	Task created but not assigned.
ASSIGNED	Task assigned to user/team or WO.
IN_PROGRESS	Field user started audit.
SUBMITTED	Field observations submitted.
DISCREPANCY_OPEN	One or more discrepancies exist.
CLOSED	Audit task completed with no open action.
CANCELLED	Task cancelled before completion.

4.3 Discrepancy Status

Status	Meaning
OPEN	Difference detected.
ROUTED	Route/action selected.
PENDING_APPROVAL	Approval required before correction.
ACTION_CREATED	Ticket, RMA, WO, or OSR action created.
RESOLVED	Discrepancy resolved.
REJECTED	Proposed discrepancy was invalid or rejected.
CLOSED	Fully closed.

5. Data Model

5.1 equipment_field_audit_campaign

What it is for: stores a planned group of field equipment audit tasks.

When to use it: create a campaign for a planned customer equipment verification, contractor quality audit, post-swap audit, area audit, or investigation batch.

Sample meaning: efac-001 is a Nairobi GPON customer equipment audit campaign covering 150 customers in August 2026.

Column	Type	Purpose	Sample
campaign_id	TEXT PK	Campaign id.	efac-001
operator_code	TEXT	Tenant/operator.	WIK
campaign_type	TEXT	CUSTOMER_EQUIPMENT_VERIFY, POST_SWAP_VERIFY, CONTRACTOR_QUALITY, AREA_INVESTIGATION.	CUSTOMER_EQUIPMENT_VERIFY
area_code	TEXT NULL	Area/region scope.	NRB-WEST
technology_family	TEXT NULL	Technology scope.	GPON
status	TEXT	Campaign status.	IN_PROGRESS
scheduled_start_at	TIMESTAMP NULL	Planned start.	2026-08-10T08:00:00Z
scheduled_end_at	TIMESTAMP NULL	Planned end.	2026-08-20T17:00:00Z
created_by_user_id	TEXT	Creator.	usr-audit-manager
created_at	TIMESTAMP	Creation timestamp.	2026-08-01T09:00:00Z
closed_at	TIMESTAMP NULL	Close timestamp.	null

Recommended indexes:

- (operator_code, status).
- (operator_code, area_code, status).
- (scheduled_start_at).

5.2 equipment_field_audit_task

What it is for: stores one field audit assignment for a customer, subscription, HomePass, site, or equipment set.

When to use it: create one task per customer premises/site/field location that must be verified by a field user or technician.

Sample meaning: efat-001 asks contractor user usr-tech-010 to verify equipment for subscription SUB-700045.

Column	Type	Purpose	Sample
audit_task_id	TEXT PK	Audit task id.	efat-001
campaign_id	TEXT NULL	Parent campaign.	efac-001
operator_code	TEXT	Tenant/operator.	WIK
task_type	TEXT	CUSTOMER_PREMISES, FIELD_SITE, POST_SWAP, INVESTIGATION.	CUSTOMER_PREMISES
customer_id	TEXT NULL	Customer id.	CUS-100045
account_id	TEXT NULL	Account id.	ACC-900045
subscription_id	TEXT NULL	Subscription id.	SUB-700045
homepass_id	TEXT NULL	Expected HomePass/site.	HP-GPON-0440
wo_id	TEXT NULL	WO if field execution is WO-backed.	wo-fa-001
assigned_to_user_id	TEXT NULL	Field user.	usr-tech-010
assigned_team_id	TEXT NULL	Contractor/team.	team-nrb-west
status	TEXT	Audit task status.	ASSIGNED
due_at	TIMESTAMP NULL	Due date/time.	2026-08-12T17:00:00Z
created_at	TIMESTAMP	Creation timestamp.	2026-08-10T08:00:00Z
submitted_at	TIMESTAMP NULL	Submission time.	null
closed_at	TIMESTAMP NULL	Close time.	null

Recommended indexes:

- (operator_code, status).
- (assigned_to_user_id, status).
- (customer_id, status).
- (subscription_id, status).
- (wo_id).

5.3 equipment_field_audit_expected_item

What it is for: stores the equipment FA-01 expects the field user to find.

When to use it: create rows from OSR-INSTANCE bindings before the task is sent to the mobile app, so the audit compares against a frozen expected state.

Sample meaning: efae-001 says the field user should find ONT instance inst-gpon-001 with serial ONT123456 at the customer site.

Column	Type	Purpose	Sample
expected_item_id	TEXT PK	Expected item row id.	efae-001
audit_task_id	TEXT	Parent audit task.	efat-001
equipment_instance_id	TEXT	OSR-INSTANCE id expected.	inst-gpon-001
sku_id	TEXT	SKU expected.	WIK-ONT-HUAWEI-EG8145V5
serial_number	TEXT NULL	Expected serial.	ONT123456
expected_location_type	TEXT	CUSTOMER_PREMISES, HOMEPASS, FIELD_SITE.	CUSTOMER_PREMISES
expected_location_ref	TEXT	Location/customer/homepass reference.	CUS-100045
expected_condition	TEXT	Expected condition.	INSTALLED_WORKING
source_snapshot_json	JSONB	OSR-INSTANCE snapshot used by the audit.	{"state": "INSTALLED", "binding": "SUB-700045"}
created_at	TIMESTAMP	Creation timestamp.	2026-08-10T08:01:00Z

Recommended indexes:

- (audit_task_id).
- (equipment_instance_id).
- (serial_number).

5.4 equipment_field_audit_observation

What it is for: stores what the field user actually found.

When to use it: create one row for each scanned/found/missing equipment item submitted by the mobile app or backoffice reviewer.

Sample meaning: efao-001 says the technician scanned serial ONT123456, confirmed it is present and working, and attached photo/GPS evidence.

Column	Type	Purpose	Sample
observation_id	TEXT PK	Observation id.	efao-001
operator_code	TEXT	Tenant/operator, used for mobile idempotency and partitioned search.	WIK
audit_task_id	TEXT	Parent audit task.	efat-001
expected_item_id	TEXT NULL	Expected item matched, if any.	efae-001
observed_equipment_instance_id	TEXT NULL	Observed instance if known.	inst-gpon-001
observed_sku_id	TEXT NULL	Observed SKU.	WIK-ONT-HUAWEI-EG8145V5
observed_serial_number	TEXT NULL	Scanned/entered serial.	ONT123456

Column	Type	Purpose	Sample
presence_status	TEXT	PRESENT, MISSING, FOUND_EXTRA, NOT_ACCESSIBLE.	PRESENT
condition_status	TEXT	WORKING, DAMAGED, TAMPERED, UNKNOWN.	WORKING
photo_file_ids_json	JSONB	Evidence file ids from file storage.	["file-001", "file-002"]
gps_latitude	DECIMAL NULL	Latitude captured by app.	-1.2921
gps_longitude	DECIMAL NULL	Longitude captured by app.	36.8219
notes	TEXT NULL	Field note.	ONT installed beside router.
captured_by_user_id	TEXT	Field user.	usr-tech-010
captured_at	TIMESTAMP	Capture timestamp.	2026-08-12T10:15:00Z
offline_client_ref	TEXT NULL	Mobile offline idempotency ref.	mob-obs-7788

Recommended indexes:

- (audit_task_id).
- (observed_serial_number).
- unique (operator_code, offline_client_ref).

5.5 equipment_field_audit_discrepancy

What it is for: stores mismatches between expected equipment state and field observation.

When to use it: create one row for missing equipment, wrong serial, found extra equipment, damaged equipment, wrong location, inaccessible site, or conflicting OSR state.

Sample meaning: efad-001 says the expected ONT serial was not found and must be routed to equipment recovery.

Column	Type	Purpose	Sample
discrepancy_id	TEXT PK	Discrepancy id.	efad-001
audit_task_id	TEXT	Parent audit task.	efat-001
expected_item_id	TEXT NULL	Expected item involved.	efae-001
observation_id	TEXT NULL	Observation involved.	efao-002
discrepancy_type	TEXT	MISSING, WRONG_SERIAL, FOUND_EXTRA, DAMAGED, WRONG_LOCATION, NOT_ACCESSIBLE.	MISSING
severity	TEXT	LOW, MEDIUM, HIGH, CRITICAL.	HIGH
status	TEXT	Discrepancy status.	ROUTED
route_action	TEXT NULL	CREATE_TICKET, CREATE_RMA_RECOVERY, REQUEST_OSR_CORRECTION, REQUEST_WRITE_OFF, NO_ACTION.	CREATE_RMA_RECOVERY
routed_ref_type	TEXT NULL	Created target type.	OSR_RMA
routed_ref_id	TEXT NULL	Created target id.	rma-001
approval_request_id	TEXT NULL	EM-CFG-04 approval request.	apr-fa-001
created_at	TIMESTAMP	Creation timestamp.	2026-08-12T10:20:00Z
resolved_at	TIMESTAMP NULL	Resolution timestamp.	null

Recommended indexes:

- (audit_task_id, status).
- (discrepancy_type, status).
- (approval_request_id).

6. APIs

6.1 Create Campaign

```
POST /api/equipment-field-audits/campaigns
Content-Type: application/json

{
  "operatorCode": "WIK",
  "campaignType": "CUSTOMER_EQUIPMENT_VERIFY",
  "areaCode": "NRB-WEST",
  "technologyFamily": "GPON",
  "scheduledStartAt": "2026-08-10T08:00:00Z",
  "scheduledEndAt": "2026-08-20T17:00:00Z"
}
```

6.2 Create Audit Task

```
POST /api/equipment-field-audits/tasks
Idempotency-Key: efat-CUS-100045-202608
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "campaignId": "efac-001",
  "taskType": "CUSTOMER_PREMISES",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "homepassId": "HP-GPON-0440",
  "assignedToUserId": "usr-tech-010",
  "dueAt": "2026-08-12T17:00:00Z",
  "createWorkOrder": true
}
```

If createWorkOrder=true, FA-01 calls WO-01 with kind=FIELD_AUDIT and stores wo_id.

6.3 Get My Audit Tasks

```
GET /api/equipment-field-audits/tasks?assignedToUserId=usr-tech-010&status=ASSIGNED
```

Used by the contractor technical mobile app.

6.4 Submit Observation

```
POST /api/equipment-field-audits/tasks/efat-001/observations
Idempotency-Key: mob-obs-7788
Content-Type: application/json
```

```
{
  "observedSerialNumber": "ONT123456",
  "observedSkuId": "WIK-ONT-HUAWEI-EG8145V5",
  "presenceStatus": "PRESENT",
  "conditionStatus": "WORKING",
  "photoFileIds": ["file-001", "file-002"],
  "gpsLatitude": -1.2921,
  "gpsLongitude": 36.8219,
  "notes": "ONT installed beside router.",
  "capturedAt": "2026-08-12T10:15:00Z",
  "offlineClientRef": "mob-obs-7788"
}
```

6.5 Submit Task

```
POST /api/equipment-field-audits/tasks/efat-001/submit
Content-Type: application/json
```

```
{
  "submittedByUserId": "usr-tech-010",
  "submittedAt": "2026-08-12T10:30:00Z"
}
```

FA-01 compares expected items and observations, creates discrepancies if needed, and updates the task status.

6.6 Route Discrepancy

```
POST /api/equipment-field-audit-discrepancies/efad-001/route
Content-Type: application/json
```

```
{
  "requestedByUserId": "usr-audit-manager",
  "routeAction": "CREATE_RMA_RECOVERY"
}
```

```
}
```

7. Rules And Drools

Use Drools for operator-variable scope, severity, and routing.

Recommended packages:

- rules.field_audit.equipment.scope
- rules.field_audit.equipment.severity
- rules.field_audit.equipment.routing
- rules.field_audit.equipment.approval

Examples:

```
package rules.field_audit.equipment.severity

rule "Missing installed customer equipment is high severity"
when
    $f : EquipmentFieldAuditFact(discrepancyType == "MISSING",
                                expectedLocationType == "CUSTOMER_PREMISES")
    $d : EquipmentFieldAuditDecision()
then
    $d.setSeverity("HIGH");
end

package rules.field_audit.equipment.routing

rule "Missing equipment creates recovery RMA"
when
    $f : EquipmentFieldAuditFact(discrepancyType == "MISSING",
                                equipmentRecoverable == true)
    $d : EquipmentFieldAuditDecision()
then
    $d.route("CREATE_RMA_RECOVERY");
end

rule "Damaged customer equipment creates support ticket"
when
    $f : EquipmentFieldAuditFact(discrepancyType == "DAMAGED")
    $d : EquipmentFieldAuditDecision()
then
    $d.route("CREATE_TICKET");
end

package rules.field_audit.equipment.approval

rule "Write-off requires approval"
when
    $f : EquipmentFieldAuditFact(routeAction == "REQUEST_WRITE_OFF")
    $d : EquipmentFieldAuditDecision()
then
    $d.requireApproval("EQUIPMENT_FIELD_AUDIT_WRITE_OFF");
end
```

Fixed validation such as allowed enum values, required evidence files, and duplicate offline references belongs in application code.

8. Events

Publish events through the transactional outbox.

Event	When emitted	Key fields
EquipmentFieldAuditCampaignCreated	Campaign created.	campaignId, campaignType
EquipmentFieldAuditTaskCreated	Task created.	auditTaskId, customerId, subscriptionId
EquipmentFieldAuditTaskAssigned	Task assigned or WO created.	auditTaskId, assignedToUserId, woId
EquipmentFieldAuditSubmitted	Field user submits observations.	auditTaskId, observationCount
EquipmentFieldAuditDiscrepancyOpened	Discrepancy detected.	discrepancyId, discrepancyType, severity
EquipmentFieldAuditDiscrepancyRouted	Route action selected/created.	routeAction, routedRefType, routedRefId

Event	When emitted	Key fields
EquipmentFieldAuditTaskClosed	Task closed.	auditTaskId, closedAt

9. Frontend And Module Consumers

Consumer	What it uses
Backoffice audit console	Campaign/task/discrepancy APIs, OSR expected state, approval status
Contractor technical mobile app	Assigned task API, observation submission, file upload, offline sync
Support/CRM	Discrepancy route to ticket or customer issue view
OSR operations	Discrepancy route to recovery, RMA, write-off, or correction
Reporting	Audit coverage, discrepancy rate, contractor quality, missing equipment value

Frontend must show expected source clearly:

- expected serial and binding come from OSR-INSTANCE snapshot stored in FA-01;
- WO state comes from WO-01 if the task is WO-backed;
- discrepancy route action comes from FA-01 and target module references.

10. Infrastructure Recommendations

For a medium deployment:

- Run FA-01 APIs in the operations/field service API deployment or a shared operations module.
- Run logical handlers in an existing operations worker deployment.
- Store evidence files in the foundation file storage service and reference only file ids in FA-01 tables.
- Support mobile offline idempotency because field technicians may submit after connectivity returns.
- Use PostgreSQL for audit/task/discrepancy state.
- Use Kafka/outbox for audit events and reporting feed.

Recommended logical worker handlers:

Handler	Purpose
fa01-generate-expected-items	Reads OSR-INSTANCE and freezes expected item snapshot.
fa01-create-wo	Creates WO-01 field audit work order when needed.
fa01-compare-observations	Compares expected items and submitted observations.
fa01-route-discrepancy	Runs Drools and creates ticket/RMA/approval actions.
fa01-close-task	Closes tasks after discrepancies are resolved or no discrepancy exists.

These are logical handlers. One worker process may implement multiple handlers.

11. Test Scenarios

Minimum tests:

1. Create audit task and freeze expected OSR-INSTANCE items.
2. Create WO when createWorkOrder=true.
3. Mobile observation submit is idempotent by offline client ref.
4. Matching serial and condition closes task with no discrepancy.
5. Missing expected serial creates MISSING discrepancy.
6. Found extra serial creates FOUND_EXTRA discrepancy.

7. Damaged equipment routes to ticket.
8. Missing recoverable equipment routes to OSR-RMA recovery.
9. Write-off route calls EM-CFG-04 approval.
10. FA-01 does not call OSR-02 procurement.

12. Implementation Notes

- Use FA-01 only for standalone audit campaigns or exception-driven verification. Do not force every normal WO completion through FA-01.
- Keep OSR-INSTANCE as the authoritative source of expected customer equipment.
- Keep WO-01 as the owner of technician assignment and field execution when the audit requires dispatch.
- Do not directly mutate equipment state from FA-01. Route corrections through the owning module.
- Design mobile APIs for offline use: deterministic client refs, duplicate-safe uploads, and resumable task submission.
- Preserve evidence and snapshots because audit findings may be reviewed weeks later.